جامعة بيرزيت
BIRZEIT UNIVERSITY

CONSTRUCTION MATERIAL

ENCE 2340

Instructor: Farhat Majadbeh

Course Objectives

The course aims to develop student's knowledge and understanding of:

- Fundamental concepts in material engineering.
- The types and characteristics of common construction materials, including concrete, steel, asphalt, and timber, which are critical for the design, analysis, and execution of construction projects.
- Concrete and asphalt mix design.
- Practical applications of construction materials, with a focus on quality assurance and control procedures.
- Environmental and sustainability considerations in the selection and use of construction materials.

Intended Learning Outcomes (ILOs)

Upon completion of this course, the student should be able to:

- Discuss the technical aspects of mechanical, thermal, and chemical properties of materials.
- Identify the properties of materials that must be considered when specifying construction products.
- Explain the environmental impact of different construction materials.
- Define and explain the fundamental properties of fresh and hardened concrete including:
    - Identify and explain the properties of concrete components including cement, aggregates, additives, and admixtures.

Intended Learning Outcomes (ILOs)

  - Analyze the behavior of fresh and hardened concrete - Evaluation of concrete behavior in both its fresh and hardened states, including workability, curing, strength development, and durability.
  - Discuss proper techniques for the mixing, transportation, handling, placing, compaction, and curing of concrete.
  - Conduct and interpret fresh and hardened concrete tests.
  - Design on a concrete mix.
- Define and explain the fundamental properties of asphalt, timber, and steel.
- Design asphalt mixes.
- Conduct and interpret test results of standard laboratory tests to determine the physical and mechanical properties of construction materials.

ILOs Concrete Technology

1. Cement :

- Identify the components of cement and describe the manufacturing processes involved in producing different types of cement.
- Discuss the chemistry of cement hydration and its impact on concrete properties.
- Identify and differentiate between various types of Portland cement and blended cement, explaining their specific purposes, properties, and applications in construction.

2. Aggregates :

- Define and classify aggregates.
- Discuss the functions of aggregates in concrete, including their role in influencing the mechanical and physical properties of the mix.
- Explain key aggregate properties, such as particle size distribution (grading), moisture content, shape, texture, specific gravity, and how these properties affect concrete performance.

3. Properties and Handling of Fresh Concrete:

- Describe the properties of fresh concrete, such as workability, and setting time.
- Explain the factors that govern these properties, including water-cement ratio, admixtures, and temperature.
- Demonstrate the correct procedures for testing fresh concrete.
- Discuss proper techniques for the transportation, handling, placing, compaction, and curing of concrete.

ILOs of Concrete Technology

4. Hardened Concrete:

- Explain the factors influencing the strength and durability of hardened concrete.
- Discuss the mechanical properties of concrete, such as compressive strength, tensile strength, modulus of elasticity.
- Conduct standard tests to assess the properties of hardened concrete (e.g., compressive strength test, split tensile test, flexural strength test) and interpret the results according to relevant standards.

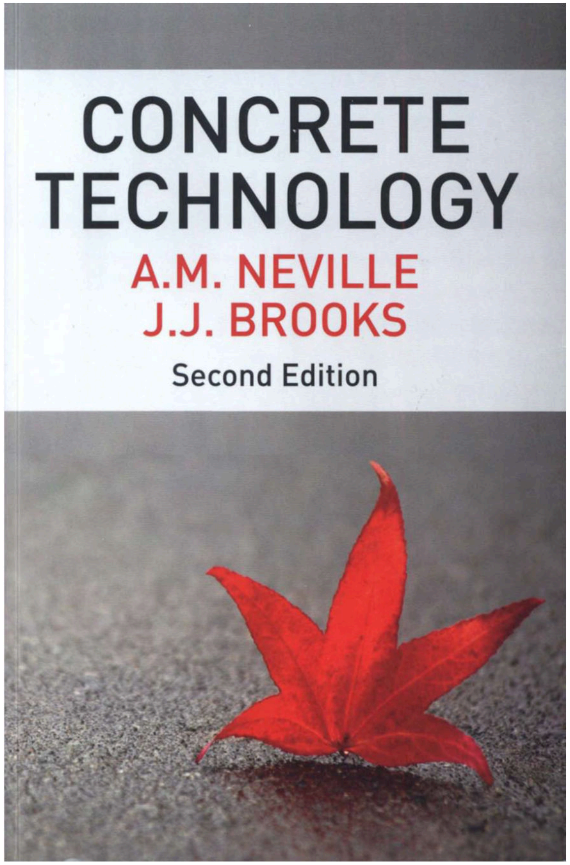
5. Concrete Admixtures and Their Effects:

- Explain the purposes and types of various concrete admixtures (e.g., water reducers, superplasticizers, accelerators, retarders, air-entraining agents) and their effects on the properties of both fresh and hardened concrete.
- Evaluate the selection of admixtures for specific applications to achieve desired concrete performance.

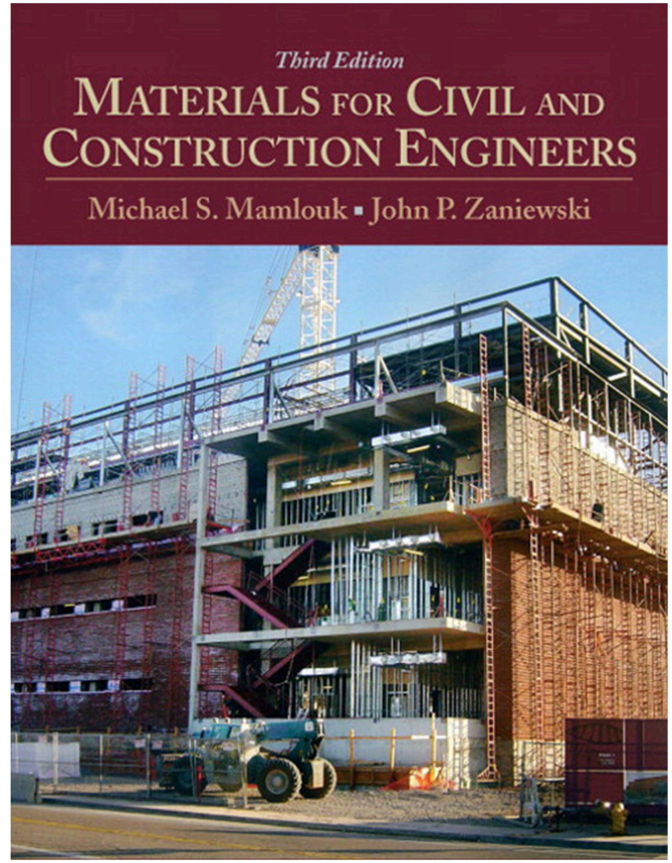
6. Concrete Mix Design:

- Design a concrete mix to achieve specific performance criteria, such as strength, workability, durability, and cost-effectiveness, considering the available materials and environmental conditions.
- Apply the principles of mix proportioning, using methods such as ACI or BS standards, and perform adjustments based on trial mixes and testing.

References



CHAPTERS 1-10, 14, 16 AND 18-20



CHAPTER 1, MATERIALS ENGINEERING CONCEPTS
CHAPTER 3,STEEL
CHAPTER 9, ASPHALT BINDERS AND ASPHALT MIXTURES
CHAPTER 10, WOOD

MATERIALS ENGINEERING CONCEPTS

Chapter 1

لمحة تاريخية – المادة والحضارة

- المواد جزء لا يتجزأ من ثقافتنا وحضارتنا. كل جانب من جوانب حياتنا اليومية تقريبًا يتأثر إلى حد ما بالمواد. (النقل، الإسكان، الملابس، الاتصالات، الترفيه، وإنتاج الغذاء).
- المواد مهمة جدًا في تطور الحضارة لدرجة أننا نربط العصور بها.
- في بداية الحياة البشرية على الأرض، استخدم الناس في العصر الحجري المواد الطبيعية فقط، مثل الحجر والطين والجلود والخشب.
- عندما اكتشف الناس النحاس وكيفية جعله أكثر صلابة من خلال خلطه بالقصدير، بدأ العصر البرونزي حوالي 3000 قبل الميلاد.
- حوالي 1200 قبل الميلاد اكتشف الحديد وهو مادة قوية أعطت ميزة في الحروب. الخطوة الكبيرة التالية كانت اكتشاف عمليات رخيصة لصنع الصلب حوالي عام 1850، مما مكن من بناء السكك الحديدية وبنية تحتية حديثة للعالم الصناعي.
- حاليا يقترن تقدم الأمم ورخائها بتقدمها في علم المواد.

لماذا ندرس المواد

بشكل أساسي، ندرس المواد لفهم خصائصها الأمر الضروري للتصميم والبناء وهذا يمكن من:

1. اختيار المادة المناسبة لاستخدام محدد بناءً على اعتبارات التكلفة والأداء.
2. فهم سلوك المواد تحت الأحمال وكيفية تغيّر خصائصها مع الاستخدام للتحقق من الأمان والاستدامة.
3. تطوير مواد جديدة ذات خصائص مرغوبة.

بشكل عام، تعد هندسة المواد خطوة أساسية نحو الابتكار.

في السباق لجعل المنشئات والأدوات والآلات أقوى وأرخص وأخف وزنًا وأكثر فعالية واستدامة، فإن التحكم بالمواد وخصائصها وعملياتها هو المفتاح.

MATERIAL GROUPS

For the purposes of discussing material properties, construction materials can be grouped into four general categories:

- Metals.
- Non-metallic inorganic materials (Ceramics).
- Polymers.
- Organic materials.

Metals

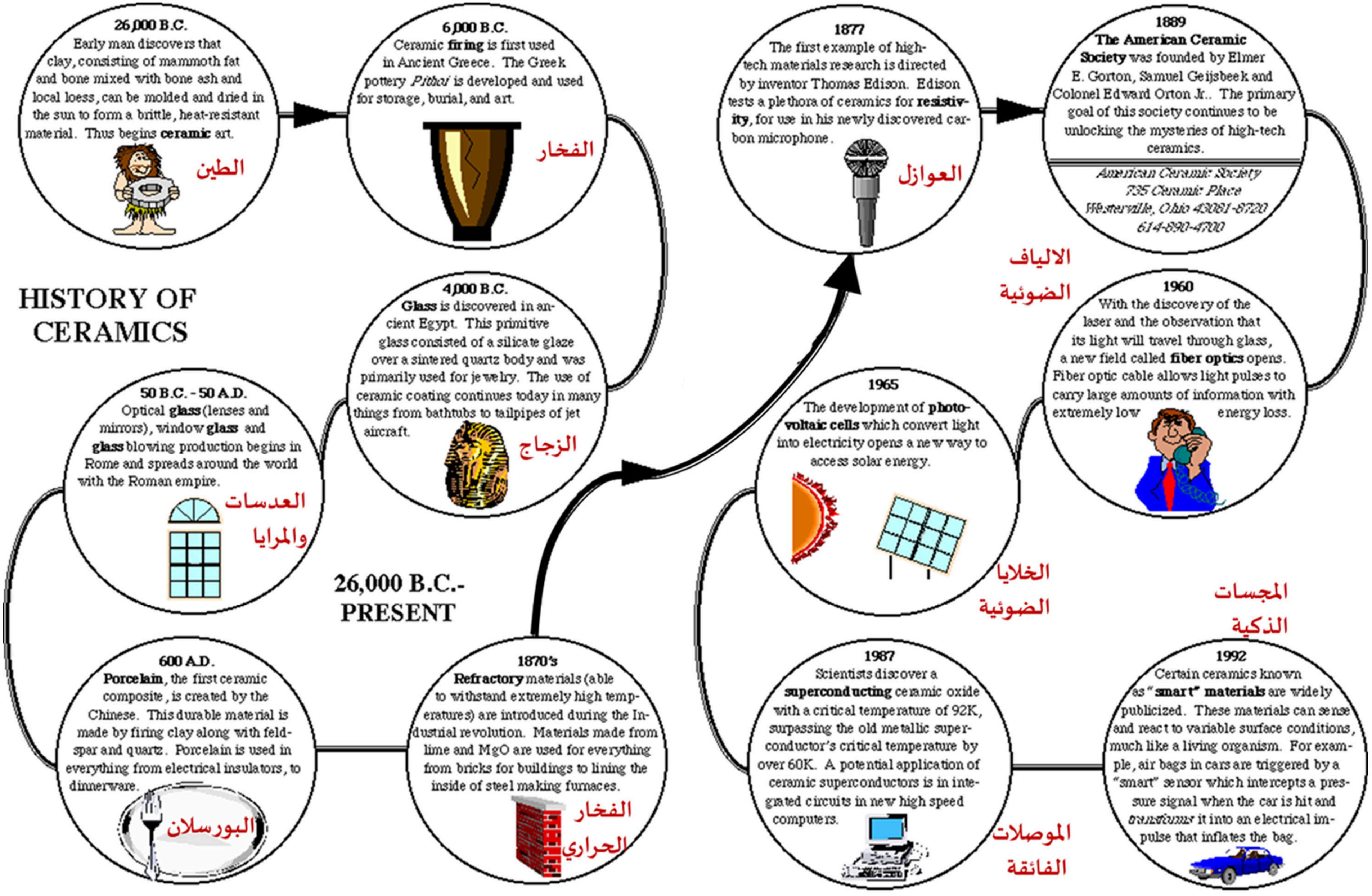
- Metals are refined from ores that have been extracted from the earth. They are composed of one or more metallic elements (such as iron, aluminum, copper), and often also nonmetallic elements (for example, carbon, nitrogen, and oxygen) in relatively small amounts.

- Metals are relatively dense, stiff and strong, yet are ductile (i.e., capable of large amounts of deformation without fracture), which accounts for their widespread use in structural applications.

Non-metallic inorganic materials (Ceramics)

- Non-metallic inorganic materials are also extracted from the soil. Often referred to as ceramics.
- They are Compounds between metallic and nonmetallic elements; most frequently oxides, nitrides, and carbides.
- Typical Ceramic materials include sand, limestone, glass, brick, cement, gypsum, mortar.
- Most are hard, rigid, brittle, and heavy, and are especially useful when compression forces are expected to occur.

Ceramics


26,000 B.C.
Early man discovers that clay, consisting of mammoth fat and bone mixed with bone ash and local loess, can be molded and dried in the sun to form a brittle, heat-resistant material. Thus begins ceramic art.
الطين
6,000 B.C.
Ceramic firing is first used in Ancient Greece. The Greek pottery Pithoi is developed and used for storage, burial, and art.
الفخار
1877
The first example of high-tech materials research is directed by inventor Thomas Edison. Edison tests a plethora of ceramics for resistivity, for use in his newly discovered carbon microphone.
العوازل
1889
The American Ceramic Society was founded by Elmer E. Gorton, Samuel Geijsbeek and Colonel Edward Orton Jr.. The primary goal of this society continues to be unlocking the mysteries of high-tech ceramics.
American Ceramic Society
735 Ceramic Place
Westerville, Ohio 43081-8720
614-890-4700
HISTORY OF CERAMICS
4,000 B.C.
Glass is discovered in ancient Egypt. This primitive glass consisted of a silicate glaze over a sintered quartz body and was primarily used for jewelry. The use of ceramic coating continues today in many things from bathtubs to tailpipes of jet aircraft.
الزجاج
الالياف الضوئية
1960
With the discovery of the laser and the observation that its light will travel through glass, a new field called fiber optics opens. Fiber optic cable allows light pulses to carry large amounts of information with extremely low energy loss.
50 B.C. - 50 A.D.
Optical glass (lenses and mirrors), window glass and glass blowing production begins in Rome and spreads around the world with the Roman empire.
العدسات والمرايا
1965
The development of photo-voltaic cells which convert light into electricity opens a new way to access solar energy.
الخلايا الضوئية
26,000 B.C.-PRESENT
المجسات الذكية
600 A.D.
Porcelain, the first ceramic composite, is created by the Chinese. This durable material is made by firing clay along with feldspar and quartz. Porcelain is used in everything from electrical insulators, to dinnerware.
البورسلان
1870's
Refractory materials (able to withstand extremely high temperatures) are introduced during the Industrial revolution. Materials made from lime and MgO are used for everything from bricks for buildings to lining the inside of steel making furnaces.
الفخار الحراري
1987
Scientists discover a superconducting ceramic oxide with a critical temperature of 92K, surpassing the old metallic superconductor's critical temperature by over 60K. A potential application of ceramic superconductors is in integrated circuits in new high speed computers.
الموصلات الفائقة
1992
Certain ceramics known as "smart" materials are widely publicized. These materials can sense and react to variable surface conditions, much like a living organism. For example, air bags in cars are triggered by a "smart" sensor which intercepts a pressure signal when the car is hit and transforms it into an electrical impulse that inflates the bag.

Polymers

- Polymers consist of large molecules composed of repeating structural units connected by chemical bonds.

- Polymers are used in a variety of construction applications, including pipes and conduit, wire and cable, textiles, adhesives, roofing, and insulation, among others
- Some of the common and familiar polymers are polyethylene (PE), nylon, (PVC).
- These materials typically have low densities, not as stiff nor as strong as metals and ceramics but very ductile.

Organic materials

- Organic materials include wood, grasses, bitumen, and many synthetic materials based on a chemical compound containing carbon.
- Examples include wood and paper products, asphalts and rubber.

Composite Materials

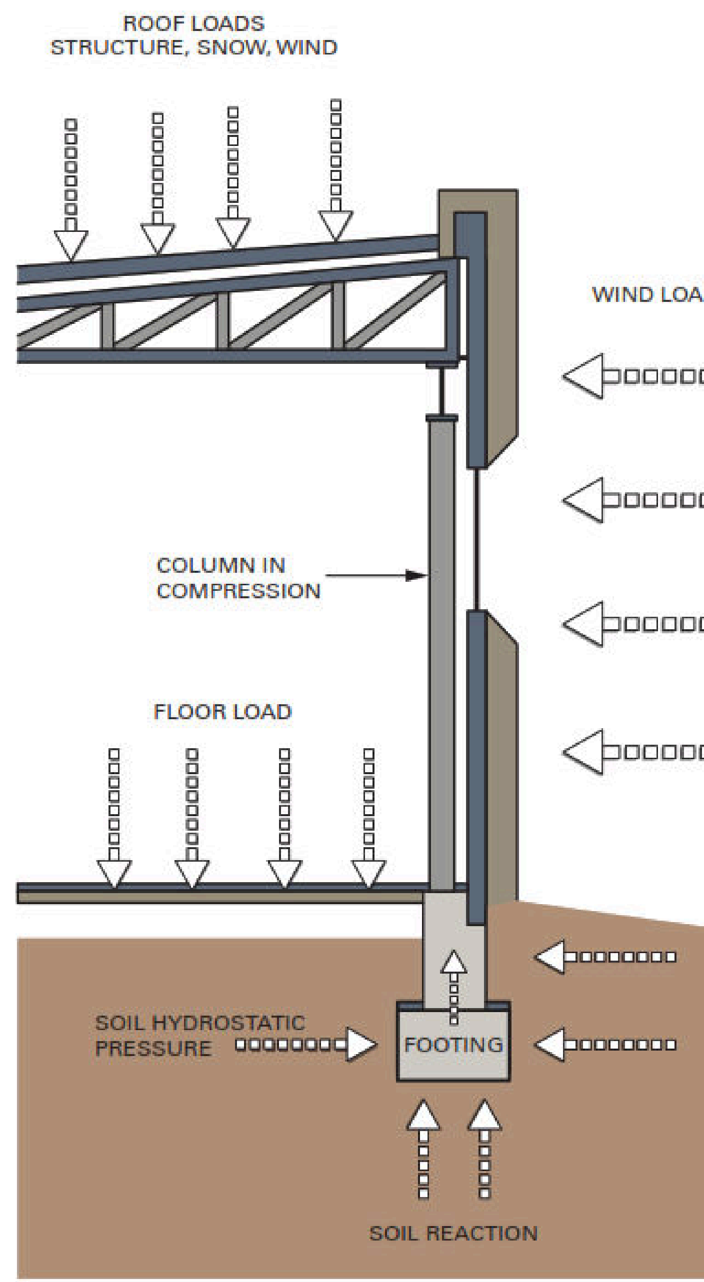
- Composite materials, or just composites, are engineered materials derived from two or more constituent materials with significantly different physical or chemical properties.
- Composites utilize the combination of two compounds: a matrix, and reinforcement. The matrix surrounds the reinforcement material, and the reinforcing imparts its special mechanical and physical properties to support the matrix properties.
- The combination results in material properties that can be engineered to produce a product or structure with optimum characteristics for a required application.
- The most visible composite today is Reinforced concrete and asphalt concrete.

Selecting Construction Materials

- The process of selecting construction materials involves careful research and evaluation to ensure they are used effectively based on their qualities and properties. Key selection criteria include:
  - ***Economic Factors***: Consider the availability and cost of materials, along with manufacturing, transportation, placement, and maintenance costs.
  - ***Mechanical Properties***: Materials must maintain their strength under various loads, stresses, and environmental conditions.
  - ***Physical Properties***: Important properties include density, thermal properties, water absorption, and surface characteristics.
- ***Fundamental new questions arise as the construction industry works to define what materials are sustainable and how to evaluate them.***

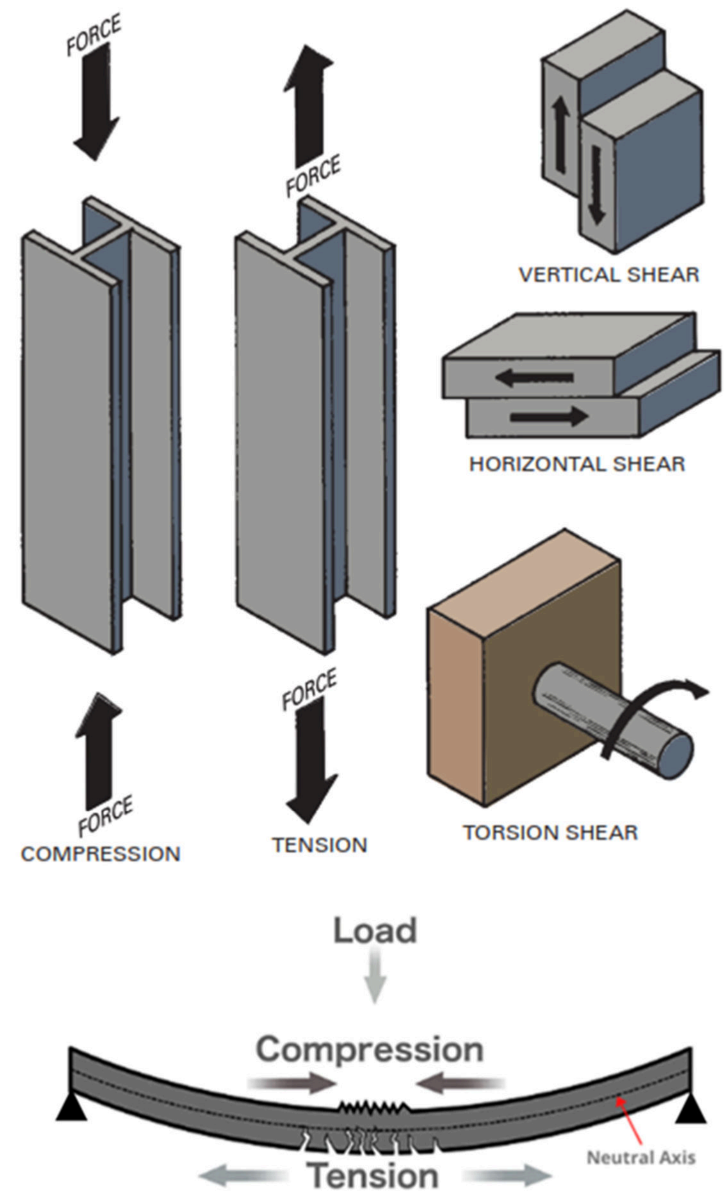
Mechanical Properties

- Mechanical properties are a measure of a material's ability to resist loads.
- Mechanical properties include tensile strength, compressive strength, shear strength, elasticity, ductility, hardness, impact resistance, and fatigue strength.



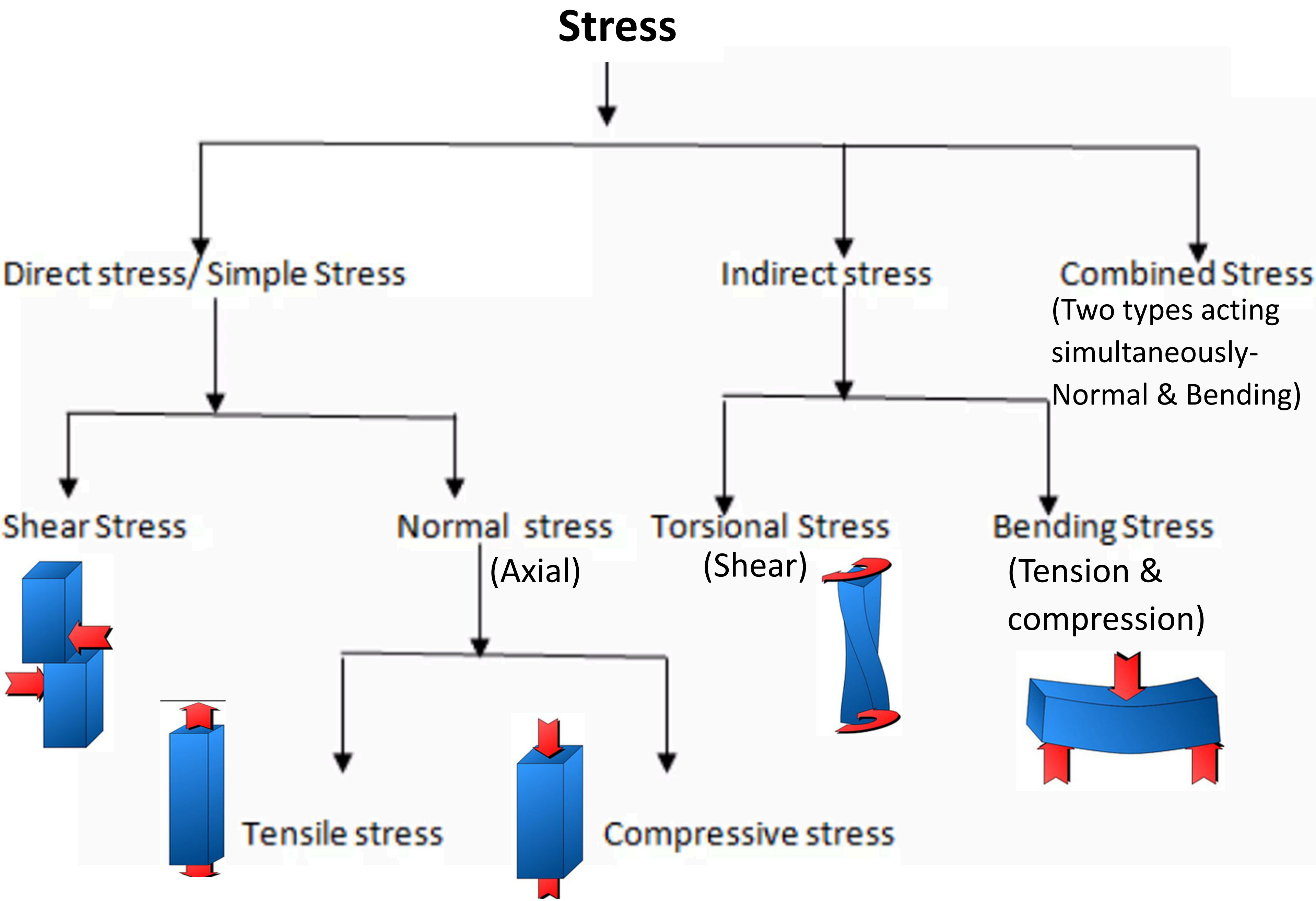
Strength of Material

- When a load, or force, acts on a building component, such as a column, the component takes on an internal resistance to the force, known as “stress.”
- As shown in the Figure, based on how loas is applied, there are many possible types of stresses that are classified as direct and indirect stress as shown in the next slide



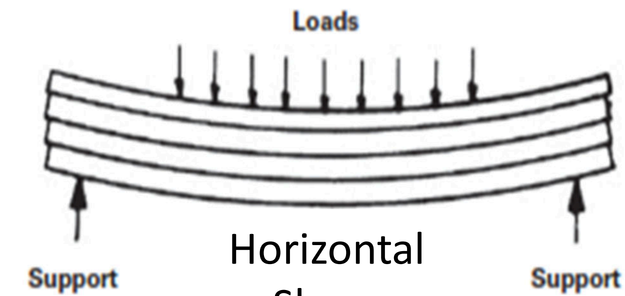
Flexural stress

Strength of Material



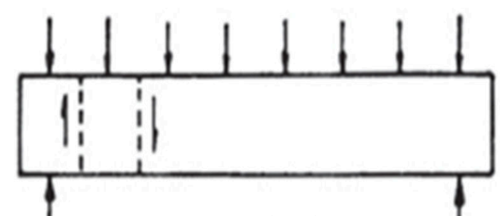
Strength of Material

1. **Tensile stresses**: Occurs when a member is subjected to a pulling or stretching force, causing elongation. This stress acts to lengthen the structural member.



Horizontal Shear

2. **Compressive stresses:** Occurs when a member is subjected to a pushing or squeezing force, causing shortening or compression. This stress acts to reduce the length of the member.

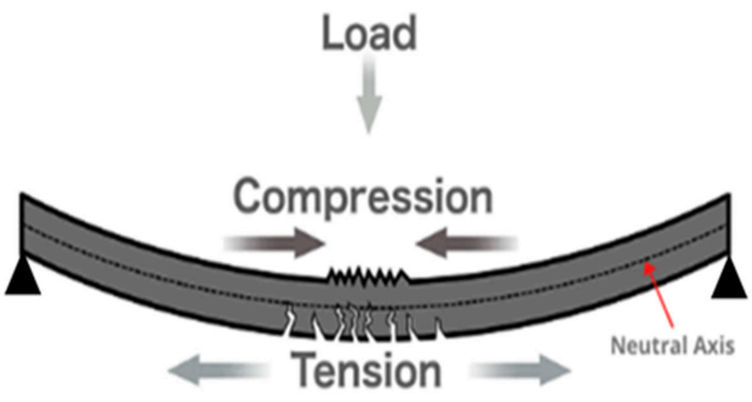


Vertical Shear

3. **Shear stresses:** Occurs when forces are applied parallel to the surface of a material, causing adjacent layers to slide relative to each other. This is common in beams and connections.

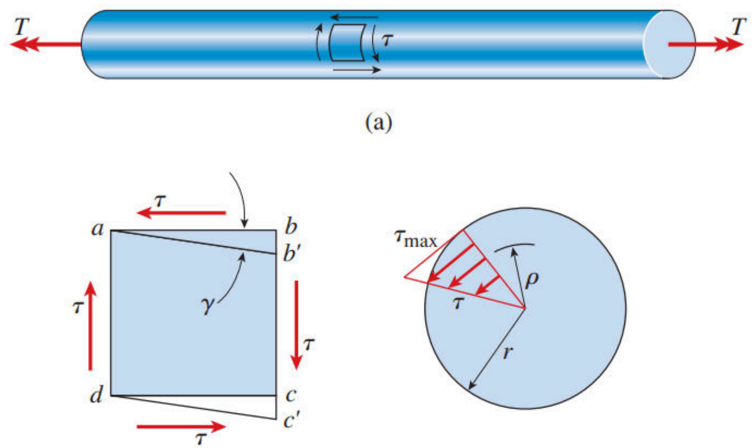
Strength of Material

4. **Flexural stresses (Bending Stress):** Occurs when a member is subjected to forces that cause it to bend. The top fibers of the member experience compression, while the bottom fibers experience tension.



Flexural stress

5. **Torsional Stress**: Occurs when a member is subjected to a twisting or rotational force, causing a twisting deformation. This is typical in shafts or beams subjected to torque.



Torsional stress

Strength of Material

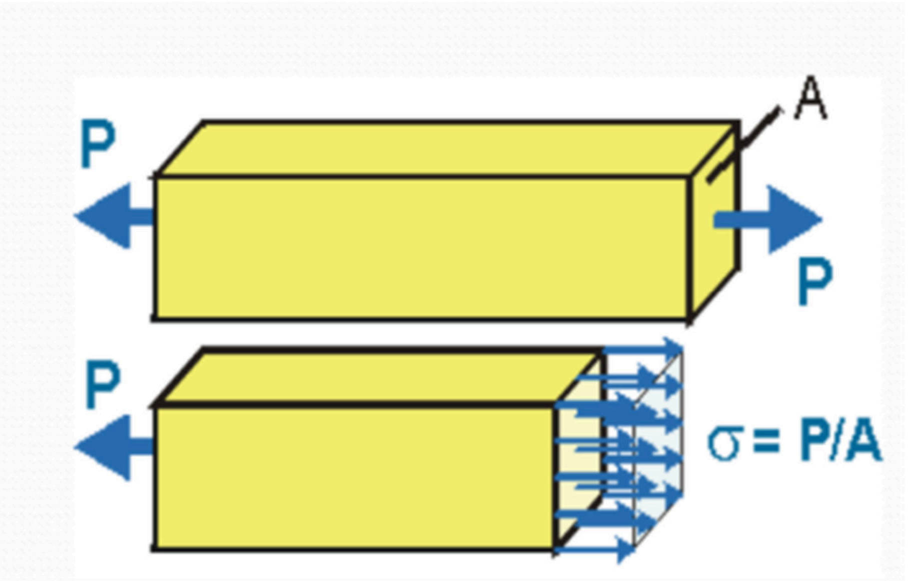
- Material Strength refers to the ability of a material to withstand applied forces or loads without failing or undergoing permanent deformation. It represents the material's resistance to breaking, bending, or deforming under stress.
- Compressive, tensile, shear, or flexural strength for a specific material, is determined through standardized tests.



Stress and Strain

- **Stress**. A measure of the internal resistance in a material to an externally applied load, measured in Newton's per square meter (Pascal's, MPa=N/mm^2).
- Normal (axial) stress is equal to the load divided by the area upon which it is acting. Normal stress caused by a force parallel to the member axis is given by:

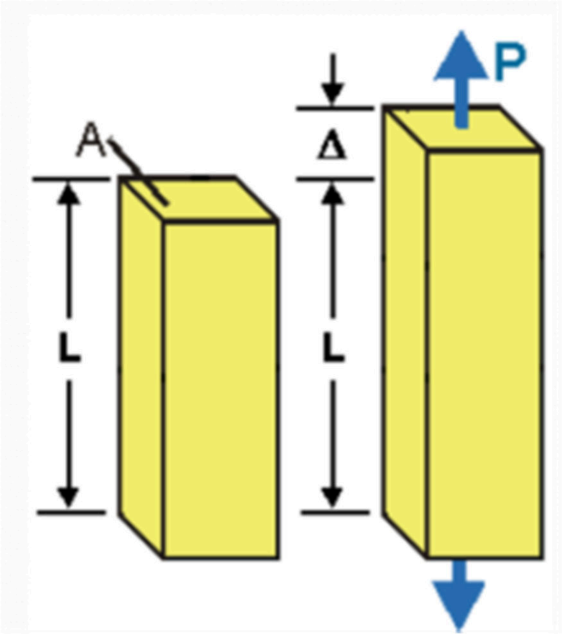
$$\sigma(stress) = \frac{Force}{Area} = \frac{P}{A}$$



Stress and Strain

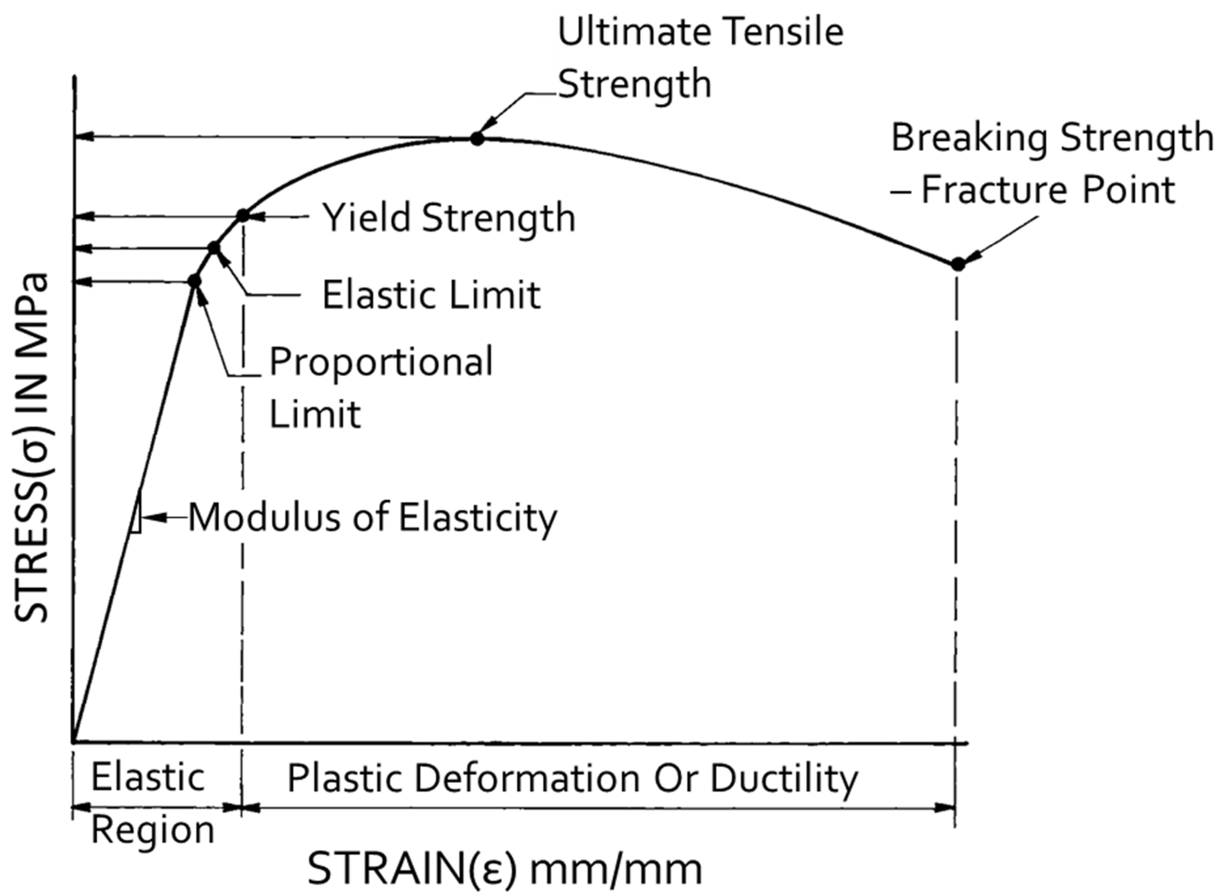
- **Strain** is the change per unit length in the linear dimension of a body that accompanies a change in stress.

$$\epsilon(strain) = \frac{Change\ in\ the\ original\ dimension}{Original\ dimension} = \frac{\Delta}{L}$$



Stress – Strain diagram

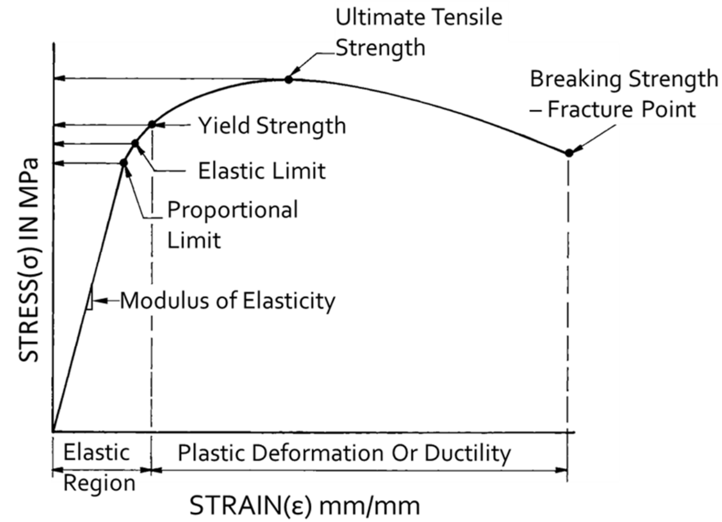
- Stress–Strain Diagram represents the behavior of a material under load. The diagram relates stress to strain for a particular material and can be obtained through standard testing.



TYPICAL STRESS–STRAIN DIAGRAM

Stress – Strain diagram

- **Elastic Region**: The initial linear portion of the curve, where the material will return to its original shape when the load is removed. The slope of this region is the **modulus of elasticity** or **Young's modulus (E)**.



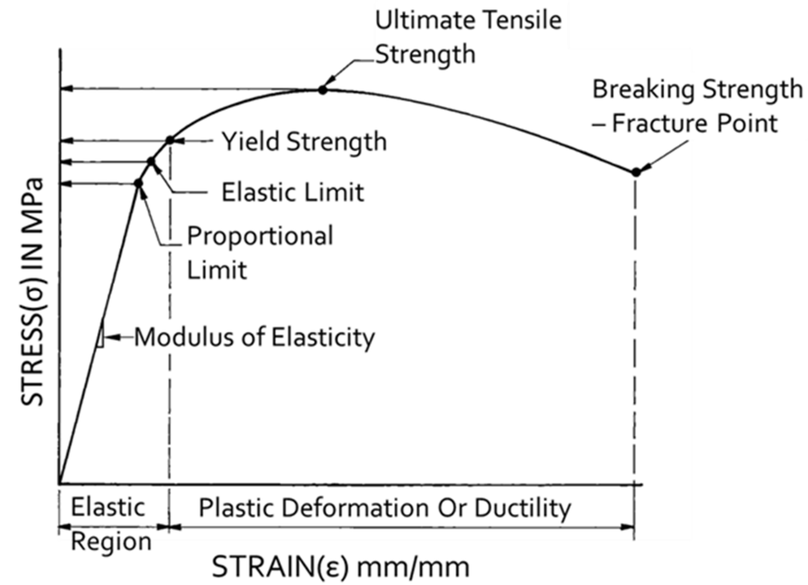
- **The modulus of elasticity** is a measure of a material's stiffness and rigidity. It determines how much a material will deflect under load and affects its buckling behavior.

- **Hook's law** $(\sigma = E\epsilon)$ **:** A fundamental principle in mechanics that relates stress to strain within the elastic range.

- **Yield Point**: The point where the material starts to deform plastically, meaning it will no longer return to its original shape once the stress is removed. This is marked by the yield strength.

Stress – Strain diagram

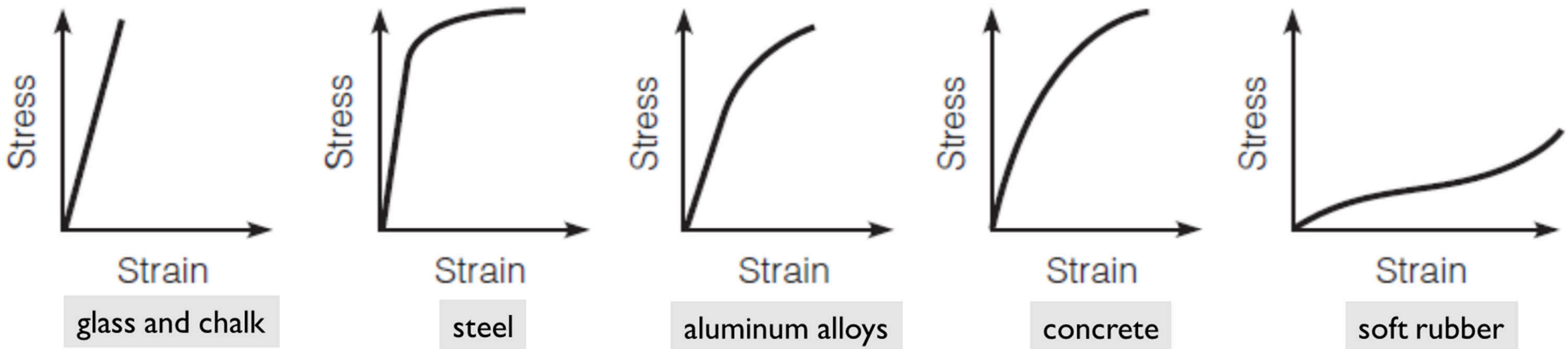
- **Plastic Region:** Beyond the yield point, the material undergoes plastic deformation. In this region, the material deforms permanently.



- **Ultimate Tensile Strength** (UTS): The maximum stress that a material can withstand while being stretched or pulled before necking. It is the highest point on the stress-strain curve.
- **Necking and Fracture**: After reaching UTS, the material begins to thin out at its weakest point (necking), and finally, it breaks or fractures. The point at which this happens is called the fracture point.

Stress – Strain diagram

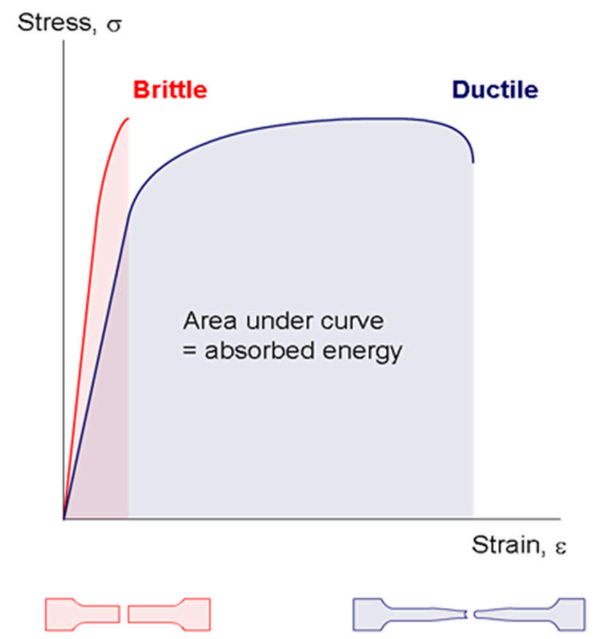
- Different engineering materials have different stress – stain diagrams.



Typical uniaxial stress–strain diagrams for some engineering materials

Ductility

- Ductility refers to the ability of a material to be deformed plastically without actually breaking or fracturing. Copper, for example, is a very ductile material. When tested under tension, it stretches for some distance without breaking.

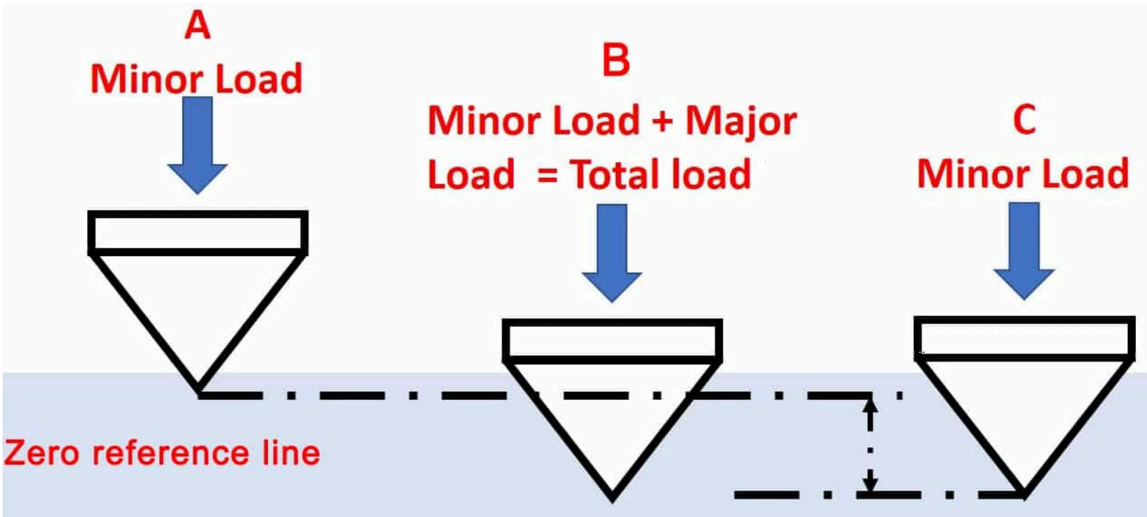


- Ductile materials are generally preferred for construction because brittle materials tend to fail catastrophically. In contrast, when a ductile material is overloaded, it may cause structural distortions, but it won't necessarily lead to a collapse. Additionally, ductile materials offer a margin of safety for designers.

Hardness

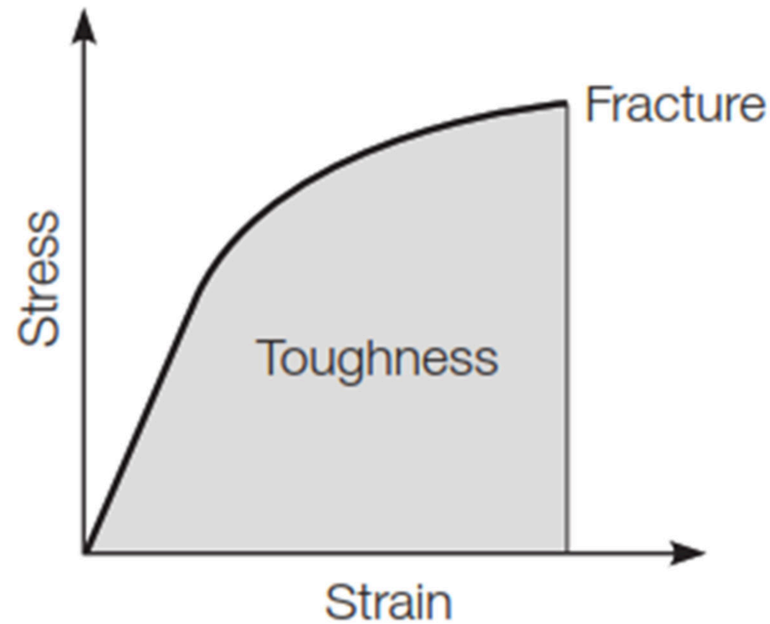
- Hardness is a measure of the ability of a material to resist indentation or surface scratching.
- It is the result of several properties of a material, such as elasticity, ductility, brittleness, and toughness.
- Hardness is related to the tensile strength of the material. Generally, a material that has a high hardness rating also will have a high tensile strength. This is not true for all materials; concrete, for example, has high hardness and low tensile strength.

Rockwell Hardness Testing



Impact strength

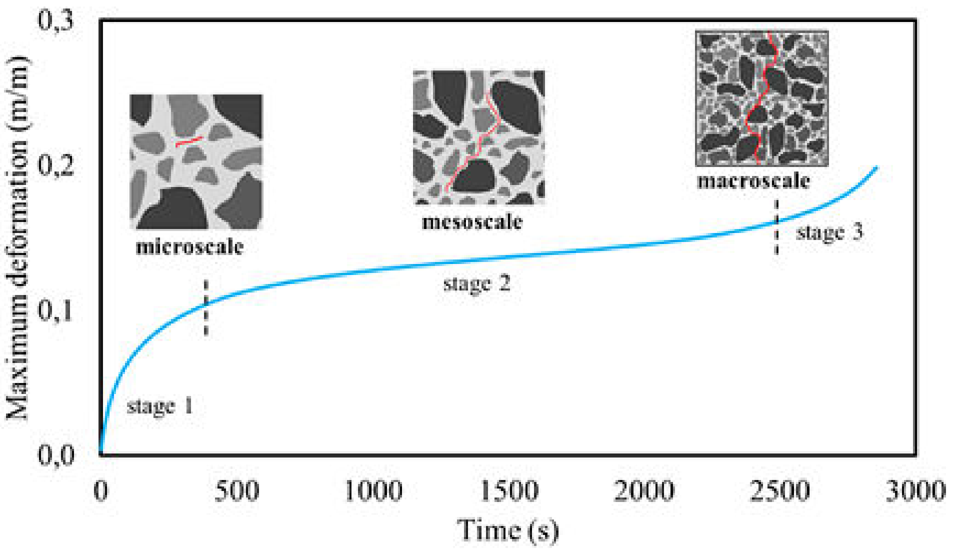
- Impact strength is the ability of a material to resist a rapidly applied load, such as the strike of a hammer.
- Impact strength is affected by strength and ductility. It is an indication of the toughness of the material.



- A material with high impact strength will absorb the energy of impact without fracturing. Metals that are strong and ductile have high impact strength. Ceramics are strong in compression but are brittle (lack ductility) and break under impact.

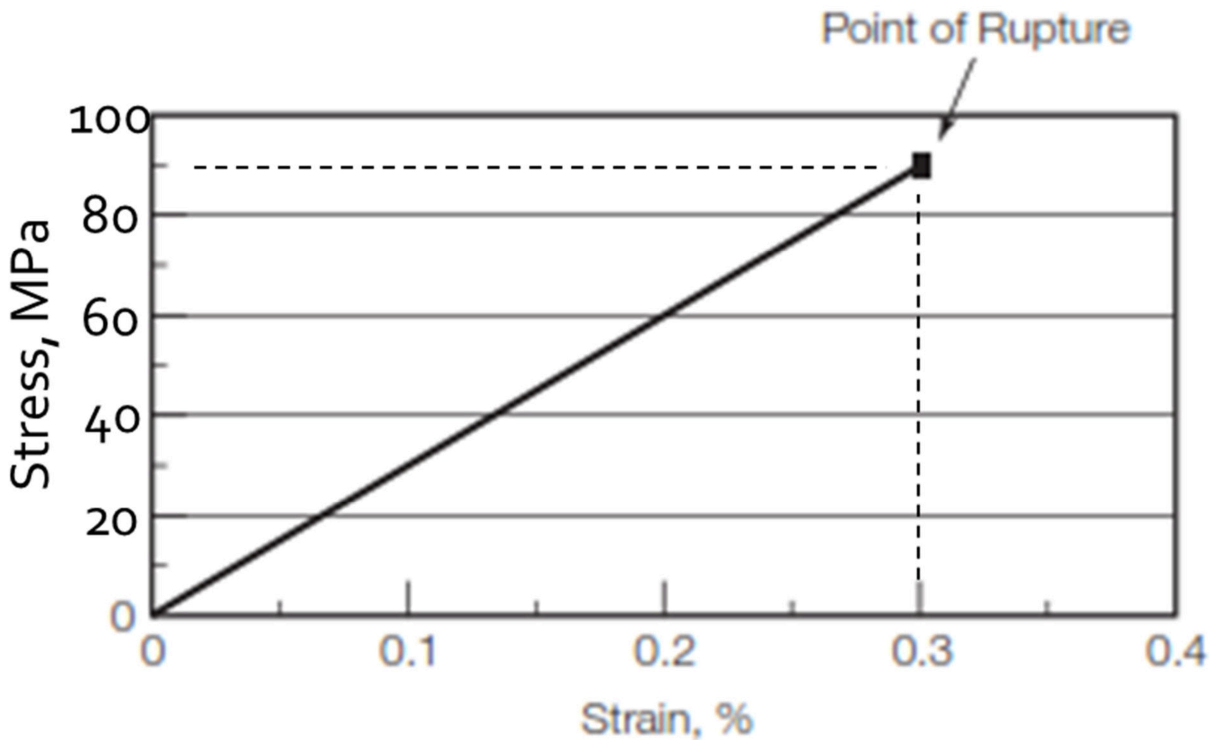
Fatigue Strength

- Fatigue strength is the resistance of a material to a cyclic load, which varies in direction and/or magnitude. This is illustrated by bending a wire back and forth until it breaks.



- Failures due to fatigue occur slowly, and most materials that fail due to fatigue offer some useful life before failure. This is an important factor to consider when the life cycle of a product is established.
- Fatigue strength in buildings and other structures, such as bridges and elevators, is commonly caused by vibrations and rotational motion produced by machinery.

Example



1. Find E of this material.
2. Find the tensile load (P) at which a bar of this material having an area of 1000 mm^2 will rupture.
3. Calculate the toughness of this material.

Material Physical properties

- Physical properties are the characteristics of the material, other than load response, that affect selection, use, and performance.
- These include:
  - Density,
  - Thermal properties,
  - Water absorption and
  - Chemical properties.

Density And Unit Weight

- The weight of the materials is an important design consideration as it determine the dead load which can significantly contributes to the total design stress. Weight is addressed through:
  - Density (ρ): the mass per unit volume of material.
  - Unit weight (ϒ): the weight per unit volume of material. ($\gamma = \rho g$)
  - Specific gravity: the ratio of the mass of a substance relative to the mass of an equal volume of water at a specified temperature. In SI units density and specific gravity terms are used interchangeably.
- For solid materials, such as metals, the unit weight, density, and specific gravity have definite numerical values. For other materials such as wood, concrete and aggregates, voids in the materials require definitions for a variety of densities and specific gravities.

Water Absorption

- It denotes the ability of the material to absorb and retain water. It is expressed as percentage in weight of the volume of dry material.
  Absorption% = [(Wet weight - Dry weight)/ Dry weight] x 100

- The properties of some building materials are greatly influenced when saturated. To identify materials that can be significantly affected by water, the ratio of the compressive strength of material saturated with water to that in the dry state shall be measured and compared. The ratio between compressive strength of material saturated to that of the dry materials is known as the **coefficient of softening**.

- Materials with coefficient of softening less than 0.8 should not be recommended in the situations permanently exposed to the action of moisture.

Thermal Properties

- The thermal properties of a material are those that are related to the material’s response to heat.
- **Thermal conductivity (κ):** denotes the ability of a material to transfer heat from an area of high temperature to an area of lower temperature. Thermal conductivity is measured in the number of BTUs (British thermal units) that can pass through one square foot of a material one inch thick when the temperature difference between the two surfaces is one degree Fahrenheit. Metals tend to have the highest values of thermal conductivity. non-metallic inorganics have much lower conductivities and organic materials have the lowest.
- **Thermal conductance (C):** is the number of BTUs per hour passing through one square foot of a material at a stated thickness in one hour per degree temperature difference.

Thermal Properties

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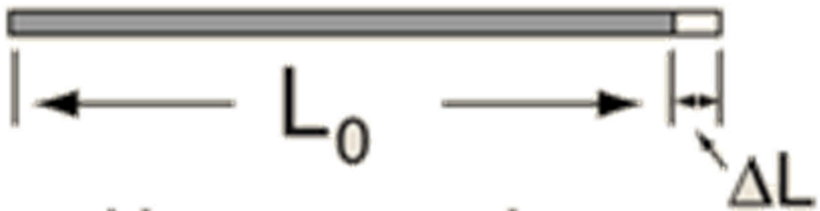
- **Thermal conductivity (κ) &Thermal conductance (C):** denotes the ability of a material to transfer heat from an area of high temperature to an area of lower temperature. Metals tend to have the highest values of thermal conductivity. non-metallic inorganics have much lower conductivities and organic materials have the lowest.
- **Thermal Expansion**: refers to the tendency of a material to change its dimensions (expand or contract) in response to changes in temperature. The amount of expansion or contraction depends on the material's coefficient of thermal expansion.

Thermal Expansion

- Coefficient of thermal expansion (α). The amount of expansion per unit length due to one unit of temperature increase.
- The relationship governing the linear expansion of a long thin rod can be reasoned out as follows:



Linear expansion

$$\Delta L = \alpha L_0 \Delta T$$

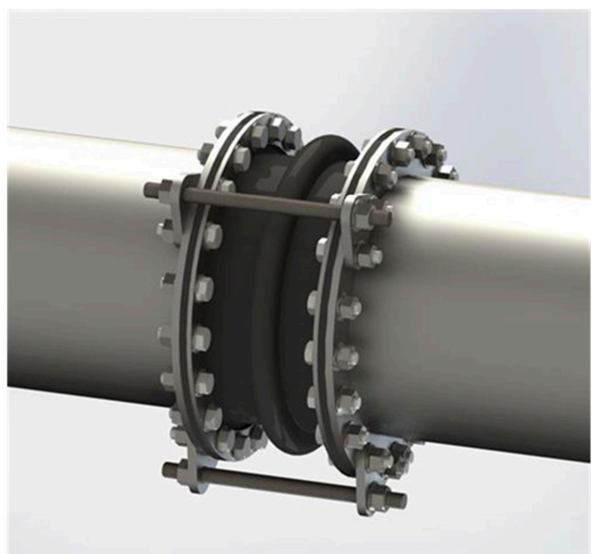
ΔT: change in temperature

Material	α (m/(m°C)
steel	0.0000117
Concrete	0.0000099

Thermal Expansion

- Understanding thermal expansion is important in engineering and construction because temperature changes can cause structural members to expand or contract, leading to stresses or deformations if not accounted for properly.
- Expansion Joints are commonly used in large structures to minimize the effects of thermal expansion and prevent damage caused by differential material expansion.

Bridge Expansion Joints



Pipe expansion Joint

Expansion Joint in Buildings

Chemical Properties

- The chemical properties of a material describe its tendency to undergo a chemical change or reaction due its composition and interaction with the environment. A chemical change can alter the original composition of a material and thereby affect its properties.

- Important chemical properties of construction materials include resistance to corrosion, ultraviolet (UV) degradation, and fire.

1. **Fire Resistance.** Combustibility is of prime importance in the selection of building materials. A material of low combustibility is said to be fire-resistant. An emphasis on the resistance to fire is a large component of building code regulations as fires create indoor air quality problems and can result in structural failure.
2. **Ultraviolet (UV) Degradation Resistance**: Ability to resist damage from prolonged exposure to sunlight.

Chemical Properties

3. Corrosion.

- The most common form of corrosion is oxidation which is the reaction of a material with the oxygen in the atmosphere, such as iron rusting.
- Other corrosive chemicals are present in the air, water, and soil to which materials are exposed. These include sulfate ions in groundwater, soils, and seawater; sulfur dioxide gas in the air; and alkali in soils.

Steel corrosion - Iron rusting

Sulfate attack in concrete

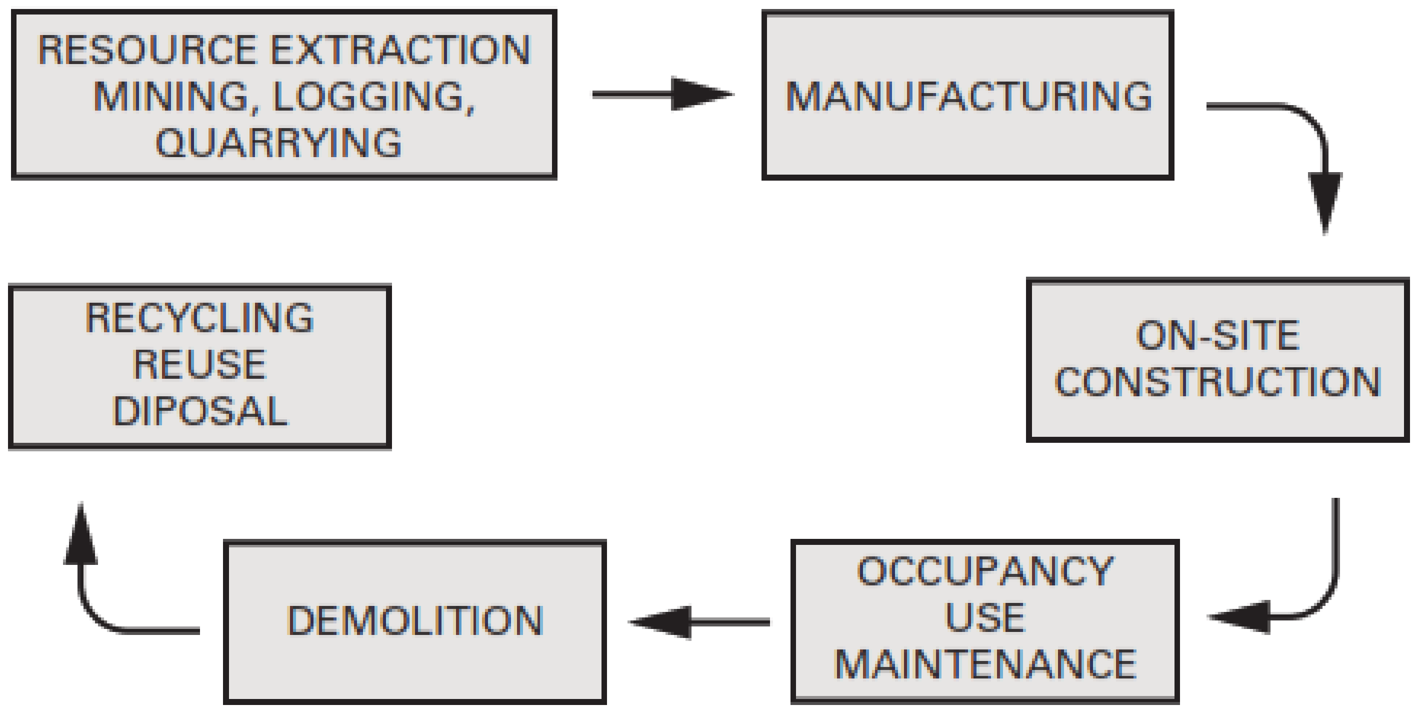
Environmentally Preferable Products (EPPs)

Sustainable Materials

- Environmentally preferable products (EPPs) or **Sustainable Construction Materials** are defined as those that have “a lesser or reduced effect on human health and the environment when compared to competing products that serve the same purpose.”
- Designers and builders should strive to specify materials that **protect the natural environment**, are **renewable and recyclable**, **minimize resource and energy consumption**, and are **healthy and nontoxic**.
- A complete life cycle assessment of proposed choices is recommended for a thorough understanding of the larger impacts of material selection.
- Life Cycle Analysis (LCA) is a comprehensive method that evaluates all resource and energy inputs, as well as emissions and waste produced throughout the entire life cycle of a material.

Embodied Energy

Embodied Energy refers to the total energy consumed throughout the entire life cycle of a construction material, from the extraction of raw resources to its manufacturing, use, and eventual reuse or demolition. Embodied energy is complex to measure and requires a full life cycle analysis (LCA).



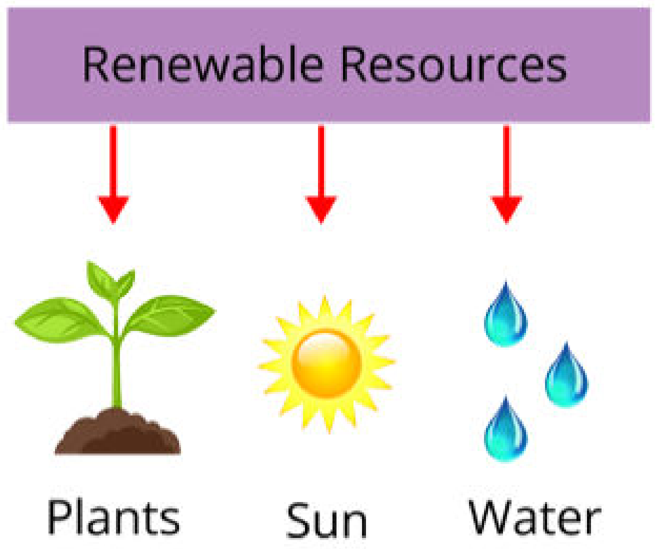
Natural Resources/Habitat Degradation

- The extraction and use of construction materials significantly impact ecosystems and natural resources.
- To minimize habitat degradation, it’s important to choose materials that reduce erosion, salinity, vegetation loss, soil nutrient changes, and landscape damage.
- Additionally, materials should be selected to avoid negative effects on water systems, such as nutrient and toxin pollution.

سلبيات صناعة الحجر في فلسطين

Renewable Materials

- Renewable materials come from living sources that can regenerate, such as plants, animals, or ecosystems. They are preferred because they can be sustained over time without depleting resources.



- Renewable materials often result in biodegradable waste and have lower CO_2 emissions throughout their lifecycle compared to fossil-fuel-based materials.
- Examples include sustainably harvested wood, bamboo, cork (الفلين), and certain composites from agricultural products. These materials are environmentally friendly, as they can clean air and water, use low-energy processes, and are recyclable.

Recycled Materials

Materials that utilize the waste products from a variety of processes results in a reduced use of natural resources and energy consumption. Recycled content is generally classified into two categories:

- Post-consumer recycled materials: These materials have been used by consumers, collected, and then reprocessed for new uses. For example, PET plastic bottles can be recycled into carpeting, fabrics, and other building materials.
- Post-industrial recycled materials: These are created from industrial waste or by-products, repurposing them into new products and keeping them out of landfills. An example is using fly ash, a by-product from coal-fired power plants, as a substitute for Portland cement in concrete.

Toxicity to the Environment

- Non-toxic building materials should be used to ensure safer indoor environments, especially since humans spend over 80% of their time indoors. Several organizations now provide certifications to label this type of materials.
- Many synthetic materials contain volatile organic compounds (VOCs) that off-gas, releasing pollutants into the air at room temperature. These pollutants, found in over 500 common building materials, can cause health problems such as asthma, allergies, cancer, and disorders of the reproductive and nervous systems.
- Materials with high contamination risks, often used in recent decades, include polyvinyl chloride (PVC), formaldehyde products like plywood and particleboard, synthetic carpets, and sheet vinyl.

Toxicity Materials